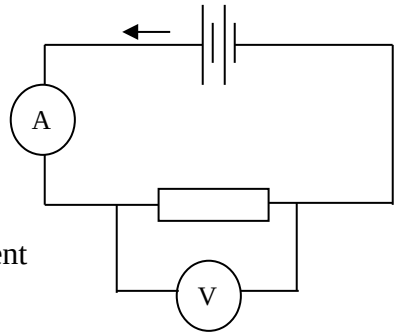
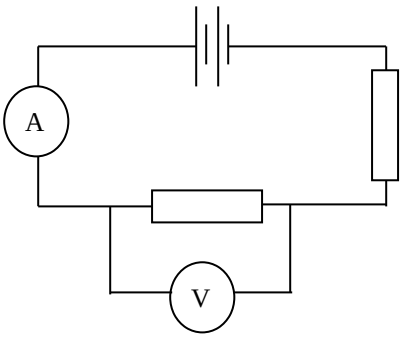
SERIES AND PARALLEL CIRCUITS

1. In a series circuit the supply voltage equals the ____***sum***_____of the voltage drops around the circuit.
2. In a series circuit the current has only one __***path***_____ to follow.
3. In a series circuit the current is _____***the same***_____ in every part of the circuit.
4. In a parallel circuit the current from the power supply is equal to the ***sum***_____of the currents in the branches.
5. In a parallel circuit the voltage drops across parallel components is _____***the same***___
6. Car headlights are wired in parallel. One advantage of this is _____***if one headlight***___ ***goes out the other one still works***_____
7. A light bulb is connected across a battery. A second light bulb is then connected in series with the first. How does this affect the brightness of the first bulb? ***It is dimmer***_____ Now the two light bulbs are rearranged so they are in parallel. How does this affect the brightness of the first bulb?__***The first bulb is back to its normal brighteness***___
8. In the circuit diagram shown;



   (a) Is this circuit series or parallel? ____***series***_____
   (b) How is the Voltmeter connected? ___***parallel***____
   (c) Draw an arrow on the wire showing which way the current is flowing.
   (d) If the battery is 3V what is the voltage reading? ___***3V***___
   (e) If the resistor is 6 Ohms what is the current reading? ***I = V/R = 3/6 = 0.5A***___

9. A second 6 Ohm resistor is added to the circuit as shown.

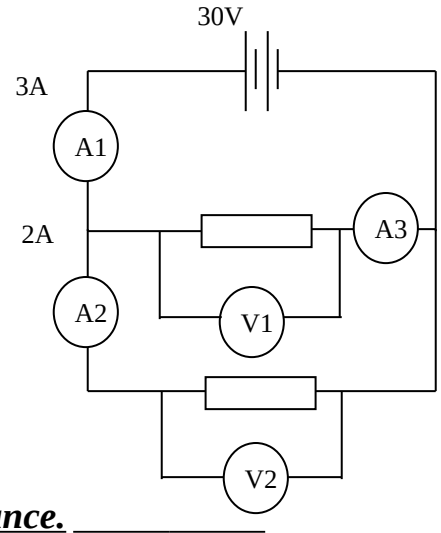


   (a) What is the total resistance of the circuit? ***12 Ohms***
   (b) What is the new voltmeter reading?__***1.5 V***_____
   (c) What is the new ammeter reading? __***0.25 A***_____
   (d) Is this a series or parallel circuit? ___***Series***_____
   (e) If one of the resistors burned out, what would happen to the current from the battery? ____***It would be zero***_____

(f) If the ammeter was shifted to between the two resistors what current value would it read? ***Same as (c) 0.25 A***

(g) If the voltmeter was shifted to be across the other resistor what voltage value would it read? ***1.5 V***

10. In the circuit shown:



(a) Is this a series or parallel circuit? ***parallel***

(b) What is the reading on A3 ***1 A***

V1 ***30 V***

V2 ***30 V***

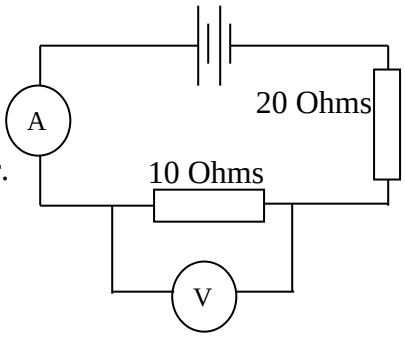
(c) Are the two resistors the same value? ***No***

Explain your answer? ***Because the resistors draw different currents from the same voltage they must have different resistance.***

(d) If the resistors were in fact light bulbs and one of the light bulbs blew out would the other light bulb go out? ***No***

Would the reading on A1 go up, or down, or stay the same? ***Down***

11. The circuit shown has two resistors. One is 10 Ohms and the other is 20 Ohms. The voltmeter reads 4 Volts.



(a) Calculate the current in the 10 Ohm resistor.
I = V/R = 4/10 = 0.4 A

(b) Calculate the voltage across the 20 Ohm resistor.
V = IR = 0.4x20 = 8 V

(c) Calculate the voltage of the battery. ***4 + 8 = 12 V***

12. Draw a circuit diagram which has a battery, a switch, a rheostat and a light bulb in series. Add an ammeter to measure the current and a voltmeter to measure the voltage across the light bulb.